Pedro Tarancón-Álvarez

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Research Interests

Cosmology	Early-universe cosmology • first-order phase transitions • bubble dynamics • gravitational waves • relativistic hydrodynamics and turbulence
QFT & holography	Strongly coupled QFTs out of equilibrium • holography (AdS/CFT) • gravitational duals from equations of state • QCD at finite temperature and density • dense matter in neutron stars
Machine learning	Physics-Informed Neural Networks for inverse and PDE problems • latent representations of PDE solution families • AI agents for scientific discovery
Methods	Numerical relativity in asymptotically AdS spacetimes • GPU simulations • spectral and finite-difference PDE solvers

Education

2023 – 2027 (exp.) **PhD in Theoretical Physics and Cosmology**, *Institute of Cosmos Sciences (ICCUB), University of Barcelona*
Thesis: *Holography in the Gravitational Wave Era* (provisional). Expected July 2027. Supervisors: Raul Jimenez and David Mateos. Holography applied to strongly coupled matter in cosmology and astrophysics: first-order phase transitions, bubble hydrodynamics and gravitational waves, with Physics-Informed Neural Networks for the associated inverse problems.

2022 – 2023 **MSc in Theoretical Physics (Particle Physics and Cosmology)**, *Autonomous University of Madrid (UAM)*, Instituto de Física Teórica UAM-CSIC, Grade: 9.33/10
Advanced QFT, Advanced Gravitation, Advanced Mathematics, with a focus on the formal aspects of quantum field theory, general relativity and quantum gravity. Thesis: *Quantum chaos in de Sitter space and Centaur geometries*, supervised by Juan F. Pedraza and Ayan K. Patra, grade 9.8/10. A 2d gravity solution that glues a de Sitter interior to an anti-de Sitter exterior, with quantum chaos diagnosed through out-of-time-ordered four-point functions computed from space-like geodesics anchored at the two AdS_2 boundaries.

2018 – 2022 **BSc in Physics**, *Complutense University of Madrid (UCM)*, Grade: 9.02/10
Honours in Algebra (10), Stellar Astrophysics (10), Differential Geometry and Tensor Calculus (9.9), Elementary Particles (9.9), Structure of Matter (9.8), Scientific Computing Laboratory (9.7), Observational Astronomy (9.5). Several courses taught in English. Thesis: *Structural relations of remote galaxies (CANDELS / 3D-HST)*, supervised by Jesús Gallego Maestro, grade 8.8/10: Python analysis of the CANDELS and 3D-HST catalogues, relating physical parameters of galaxies over the redshift range 0.4–1.5.

Research Positions and Visits

2026 **Research visit (forthcoming)**, *Institute for Advanced Study, Princeton*, Host: J. M. Stone, Dec.
Effective theory of vortex interactions, based on enstrophy and a multipole expansion.

2023 – present **PhD Researcher, FPI Fellow**, *ICCUB, University of Barcelona*, Supervisors: R. Jimenez & D. Mateos
Predoctoral fellowship from the Spanish Ministry of Science and Innovation, as part of the project *Dark Energy and the Origin of the Universe*. Holographic description of strongly coupled quantum fields in cosmology and astrophysics (phase transitions in the early universe and in neutron-star mergers), with Physics-Informed Neural Networks for the associated non-linear inverse problems.

2025 **Research visit, Center for Computational Astrophysics**, *Flatiron Institute, New York*, Host: F. Villaescusa-Navarro, June
Work on the *Denario* multi-agent AI system; set up access to the Rusty cluster.

2024 **Visiting Researcher, StellarDNN Lab**, *School of Engineering and Applied Sciences, Harvard University*, Host: P. Protopapas, Apr–Jul
Physics-Informed Neural Network methods for holographic and cosmological inverse problems; the start of an ongoing series of joint papers.

2023 **JAE Intro Research Fellow (CSIC)**, *Instituto de Física Teórica UAM-CSIC, Madrid*, Supervisor: J. F. Pedraza, Mar–Sep
Holography and quantum chaos: out-of-time-order correlators in Centaur geometries, which connect an AdS exterior to a dS interior.

2021 – 2022 **Undergraduate Research Fellow (Ministry of Education)**, *Earth Physics and Astrophysics Dept., UCM*
Prototype control system for the MOSAIC instrument of the ELT, within the GUAIX and ISCAR groups (Profs. África Castillo and Jesús Gallego).

Collaborations

2025 – present Member of the **Simons Initiative on the Geometry of Flows**.

Fellowships and Awards

2023 – 2027 **FPI Predoctoral Fellowship**, Spanish Ministry of Science and Innovation (PRE2022-102220), awarded through national competition.

2023 **JAE Intro Fellowship**, Spanish National Research Council (CSIC), awarded through national competition.

2021 – 2022 **Beca de Excelencia** (Excellence Scholarship), Comunidad de Madrid, for outstanding academic performance.

2021 – 2022 **Collaboration Grant**, Spanish Ministry of Education, for undergraduate research (UCM).

Publications

10 papers: 2 in *JHEP* (one published, one to appear), 1 in *Communications Physics* (Nature Portfolio), 1 in *PRX Intelligence*, 1 in the ICLR 2026 AI & PDE Workshop proceedings, and 5 preprints (3 under review). Lead author on 6.

Note on author ordering. The author lists of **P2**, **P6**, **P7** and **P10** are alphabetical. **P1** and **P4** list a first group of authors followed by an alphabetical list. In **P3**, **P5**, **P8** and **P9** the ordering reflects contribution. My own contribution is stated under each entry. Lead author on **P1**, **P2**, **P3**, **P5**, **P6** and **P7**.

Journal articles

P1 Y. Bea, R. Jimenez, D. Mateos, S. Liu, P. Protopapas, **P. Tarancón-Álvarez**, P. Tejerina-Pérez,
"Gravitational duals from equations of state",
Journal of High Energy Physics **07** (2024) 087. doi:10.1007/JHEP07(2024)087 · arXiv:2403.14763.
Holography as an inverse problem, in which the five-dimensional dual of a strongly coupled gauge theory is reconstructed from its equation of state alone, using Physics-Informed Neural Networks. *My contribution:* leading author; designed, implemented and trained the PINN solver; numerical experiments; writing.

P2 R. Jimenez, D. Mateos, P. Protopapas, P. Solé-Vilaró, **P. Tarancón-Álvarez**, P. Tejerina-Pérez,
"Gravitational Duals from Equations of State II: Large Hierarchies and False Vacua",
To appear in *Journal of High Energy Physics*. arXiv:2606.30117.
Extension of **P1** to large scale hierarchies and to equations of state with a false vacuum, recovering parts of the bulk potential not directly encoded in the thermodynamic data. A step towards computing the bubble dynamics, and ultimately the gravitational-wave spectrum, of a given strongly coupled theory. *My contribution:* leading author; the machine-learning methodology and its adaptation to stiff and hierarchical regimes; numerical experiments; writing.

P3 **P. Tarancón-Álvarez**, P. Tejerina-Pérez, R. Jimenez, P. Protopapas,
"Efficient PINNs via multi-head unimodular regularization of the solutions space",
Communications Physics (Nature Portfolio) **8** (2025) 335. doi:10.1038/s42005-025-02248-1 · arXiv:2501.12116.
A multi-head PINN architecture with a unimodular regularization of the latent space, which substantially improves transfer learning into stiff regimes. *My contribution:* first author; conception of the method, theoretical formulation, implementation, experiments and writing.

P4 F. Villaescusa-Navarro, B. Bolliet, P. Villanueva-Domingo, A. E. Bayer *et al.* (36 authors, including **P. Tarancón-Álvarez**),
"Denario Project: Deep Knowledge Artificial Intelligence Agents for Scientific Discovery",
PRX Intelligence **1** (2026) 021001. doi:10.1103/kk55-gc95 · arXiv:2510.26887.
A modular multi-agent system for research tasks: idea generation, literature review, code writing, analysis and paper drafting. *My contribution:* physics-domain test cases; writing. Denario proposed the principal-component analysis of neural embeddings later developed in **P7**.

Refereed workshop proceedings

P5 **P. Tarancón-Álvarez**, L. Sarieddine, P. Protopapas, R. Jimenez,
"Learning embeddings of non-linear PDEs: the Burgers' equation",
AI & PDE: ICLR 2026 Workshop on AI and Partial Differential Equations, Rio de Janeiro. arXiv:2603.07812.
Neural solvers of the viscous Burgers equation organise their latent space onto a low-dimensional solution manifold. *My contribution:* first author; method, experiments and writing. Presented as a poster at the workshop.

Preprints

P6 D. Mateos, M. Sanchez-Garitaonandia, **P. Tarancón-Álvarez**,
"Non-conformal obstructions to bubble expansion",
Submitted (2026). arXiv:2607.27874.
New hydrodynamic obstructions to bubble expansion for non-conformal equations of state, together with a new branch of solutions, the shocked detonations. For QCD-like equations of state all detonations are excluded, which lowers the kinetic energy available for gravitational-wave production. *My contribution:* leading author; led the numerical programme; wrote the public bubble solver `SNOBEX`; the scan over equations of state and the kinetic-energy budget analysis; writing.

P7 R. Jimenez, S. Mayboroda, P. Protopapas, L. Sarieddine, D. N. Spergel,
P. Tarancón-Álvarez,
"Physics-Informed Neural Embeddings of PDE Solution Families",
Submitted (2026). arXiv:2607.06348.
Finite-dimensional embeddings of *families* of PDE solutions. For Burgers dynamics, two to four principal components capture ∼95% of the latent variance, and their Fourier-shell profiles are invariant under the residual basis freedom of the latent space, and therefore reproducible across independent training runs. *My contribution:* leading author; led the design of the PINN architecture, the implementation, and the Fourier-shell decomposition; much of the writing.

P8 A. A. Kalout, P. Tejerina-Pérez, K. Karchev, **P. Tarancón-Álvarez**, L. Sarieddine,
R. Jimenez, M. Engelstein, G. David,
"Recovering Sharp Conductivity Features in the Finite-Data Calderón Problem with Physics-Informed Neural Networks",
Submitted (2026). arXiv:2606.28158.
Neural inversion for the ill-posed Calderón problem from finite Dirichlet-to-Neumann data, using randomised multiscale wavelet excitations and Fourier-feature encodings. Dominant conductivity structure is recovered to 3–12% relative error. *My contribution:* conceptual development and design of the PINN architecture; the Fourier-feature study and the multiscale boundary excitations; writing.

P9 R. Jimenez, P. Tejerina-Pérez, L. Sarieddine, **P. Tarancón-Álvarez**, M. Traforetti,
P. Solé-Vilaró, O. Ignatenko, N. Tkach, P. Protopapas,
"Low Residuals, Wrong Value: Learning the Viscosity Solution of Hamilton–Jacobi–Isaacs Equations with PINNs",
Preprint (2026). arXiv:2610.27193.
For Hamilton–Jacobi–Isaacs equations, a small PDE residual does not validate a learned solution: standard PINNs converge to almost-everywhere solutions that differ from the viscosity solution by order one. Feeding the network geodesic-distance information recovers the viscosity solution, and a mesh-free semiconcavity test flags the failure without the exact solution. *My contribution:* structure of the paper; scientific discussion; writing of the final draft.

P10 A. Caddeo, O. Henriksson, C. Hoyos, M. Sanchez-Garitaonandia, **P. Tarancón-Álvarez**,
"Effective scalar–fluid description of thermal first-order phase transitions and their spinodal instabilities",
Preprint (2026). arXiv:2611.13102.
An improved scalar–fluid effective description of first-order phase transitions in relativistic theories, tested against a microscopic calculation in a holographic model, with a friction coefficient that governs the bubble wall velocity and can be extracted from a holographic correlator. *My contribution:* developed the `Julia` code (`QUASTELL`, **S5**); analysis and interpretation of the results for varying backreaction; organisation of the code repository; writing and revision of the manuscript.

Software

S1 SNOBEX (sole author): the complete space of self-similar bubble solutions for an arbitrary equation of state, including the new shocked-detonation branch. Released with **P6**.

S2 NNHolo: PINN solver for the holographic inverse problem, behind **P1** and **P2**.

S3 Unimodular_regularization: multihead PINN with unimodular regularization of the latent space, released with **P3**.

S4 PDE_embeddings (sole author): multihead PINN framework and latent-space diagnostics for PDE solution manifolds (**P5**, **P7**), built on `Neurodiffeq`.

S5 QUASTELL: `Julia` package for the backgrounds, thermodynamics and quasinormal-mode spectra of holographic black branes, released with **P10**.

S6 Contributor to AthenaK, the performance-portable version of the `Athena++` framework: implementation of the turbulence forcing.

S7 P2_ML_Lectures_public: lecture notes and exercises for the Master's course.

Talks and Posters

2026 *Non-conformal obstructions to bubble expansion*, talk, Barcelona Initiative for Gravitation (BIG) meeting, Barcelona, Oct. 30. [slides]

2026 *Non-conformal obstructions to bubble expansion*, talk, workshop "VAST 2026: From Vacuum Stability to Gravitational Waves", Madrid, Oct. 21. [slides]

2026 *Learning embeddings of non-linear PDEs: the Burgers' equation*, poster, AI & PDE: ICLR 2026 Workshop, Rio de Janeiro, Apr. [poster]

2026 *Non-conformal obstructions to bubble expansion*, poster, Iberian Strings 2026, IGFAE, Santiago de Compostela, Jan. [poster]

2025 *The Denario Project: Deep Knowledge Agents for Scientific Discovery*, oral, Cross-Collserola PhD Meeting, IFAE, Universitat Autònoma de Barcelona, Oct. [slides]

2025 *The Denario Project*, internal seminar (with L. Verde and R. Jimenez), ICCUB, University of Barcelona, Sep. [slides]

2025 *Gravitational Duals from Equations of State*, poster, conference "Gravity: New Quantum and String Perspectives", Centro de Ciencias de Benasque Pedro Pascual, Jul. [poster]

2025 *Gravitational Duals from Equations of State*, **invited talk**, workshop "New Insights in Black Hole Physics from Holography", Instituto de Física Teórica (IFT), Madrid, Jun. [slides]

2025 *Gravitational Duals from Equations of State*, seminar, Theoretical Physics Department, University of Florence, May. [slides]

2025 *Gravitational duals from equations of state*, gong show, Iberian Strings 2025, Bilbao, Jan. [slides]

2024 *Gravitational duals from equations of state*, seminar, Gran Sasso Science Institute, L'Aquila, Oct. [slides]

Teaching

2025 – 2027 **Guest Lecturer, "Mathematical and Statistical Techniques"**, *MSc in Astrophysics, Particle Physics and Cosmology, University of Barcelona*
Lecture block in English as part of the core methods course of the Master's programme (42 students in 2025–26). Design and delivery of the lectures and exercises on *machine learning, Physics-Informed Neural Networks and dimensionality reduction*; materials released through the course GitHub repository (**S7**).

Professional Service

2026 Referee for *Astronomy and Computing* (Elsevier); certificate of reviewing awarded.

2025 – present Organiser of the seminars of the Quantum Field Theory and String Theory group, ICCUB, University of Barcelona.

Schools, Workshops and Training

2026 Workshop on Holography and Dense Matter, APC, Paris.

2026 Geometry of Flows meeting, New York University, New York.

2026 Simons Initiative on the Geometry of Flows retreat "Explainability in ML & Multi-Agent Inference" (mathematical hydrodynamics and turbulence), Canfranc Underground Laboratory.

2026 Geometry of Flows Workshop (geometry of fluid flows, hydrodynamics and turbulence), Flatiron Institute, New York.

2026 AI & PDE: ICLR 2026 Workshop on AI and Partial Differential Equations, Rio de Janeiro.

2026 Iberian Strings 2026, IGFAE, Santiago de Compostela.

2025 ETH-ITS Workshop "Geometry of Flows" (hydrodynamics and turbulence), ETH Zurich.

2025 Simons Initiative on the Geometry of Flows meeting (mathematical hydrodynamics and turbulence), Institute for Advanced Study, Princeton.

2025 "Gravity: New Quantum and String Perspectives", Centro de Ciencias de Benasque Pedro Pascual.

2025 Workshop "New Insights in Black Hole Physics from Holography", IFT Madrid.

2025 GGI School on asymptotic symmetries and flat holography, Galileo Galilei Institute, Florence.

2025 Iberian Strings 2025, Bizkaia Aretoa, Bilbao.

2024 – 2026 Barcelona Initiative for Gravitation (BIG) meetings, Barcelona: from the 4th meeting (2024) to the October 2026 meeting.

2023, 2025 Cross-Collserola PhD Meeting in Astrophysics, Cosmology and Particles, Barcelona.

2024 "The Complexity of the Cosmos" and "Gravitational duals from equations of state", Gran Sasso Science Institute, L'Aquila.

2024 COSMO CLASSIC 2024: future prospects for Cosmology, Asiago.

2024 New Frontiers in Strong Gravity, Centro de Ciencias de Benasque Pedro Pascual.

2024 GGI School on Theoretical Aspects of Astroparticle Physics, Cosmology and Gravitation, Florence (60 h).

2024 CERN Winter School on Supergravity, Strings and Gauge Theory, Geneva.

2024 ComputeFest 2024: Physics-Informed Neural Networks, Harvard IACS (16 h).

2023 – 2024	ICCUB Machine Learning Workshop (2023) and ICCUB Winter Meeting (2024), Barcelona.
2021	Phy6cool summer school on particle physics and cosmology, CIEMAT, Madrid (47 h).

Technical Skills

Numerical	Numerical relativity in asymptotically AdS spacetimes; relativistic hydrodynamics; turbulence forcing in *AthenaK*; spectral and finite-difference PDE solvers; HPC clusters (including the Flatiron Rusty cluster).
Machine Learning	Physics-Informed Neural Networks, multihead architectures and transfer learning, deep learning for inverse and PDE problems, representation and manifold learning, dimensionality reduction. PyTorch, JAX.
Programming	Advanced Python and Julia; Wolfram Mathematica (advanced); Matlab (intermediate); C++ (basic, turbulence forcing in the *AthenaK* code); LaTeX (advanced); Git and version control.
Physics	AdS/CFT and holography, quantum field theory, general relativity, numerical relativity in asymptotically AdS spacetimes, relativistic hydrodynamics, cosmology.
Languages	Spanish (native). English (fluent; working language of research and teaching; B2, EOI Raimundo de Toledo). Catalan (basic). French (basic; A2, EOI Raimundo de Toledo).

Outreach and Media

2025	Research (lead author) covered in the Spanish national press: *La Vanguardia*, "UB researchers develop new AI techniques to solve complex equations in physics" (Oct. 2025) [link].
2025	Video presentation "The Denario Project: Modular Automation of Scientific Research with Multi-Agent Systems" [YouTube].
2022	Co-organiser and speaker, seminar on the MOSAIC instrument for the ELT, X Jornadas de Introducción a la Investigación, UCM.
2021	Volunteer, citizen-science activity *StreetSpectra* (spectra of street lights), Science Week, UCM.
2020	**Winner** (video category), outreach contest *Luz en Casa*, International Day of Light 2020, Department of Optics, UCM, with the team video *Experimento de Hertz*.

Other Interests

Amateur astronomy (telescope operation and calibration); piano (professional-level education, Conservatorio Profesional de Música Jacinto Guerrero); reading and music.